Power BI Assignment

Q1. What is Power BI and how does it differ from Excel?

Ans:

Power BI is a business intelligence tool developed by Microsoft for data visualization and creating interactive reports. While Excel is primarily a spreadsheet tool for organizing and calculating data, Power BI is designed specifically for advanced analytics.

Key differences between Excel and Power BI:

- **Data Handling:** Excel is suitable for small to medium datasets, whereas Power BI can handle very large datasets easily.
- **Visualization:** Excel offers basic charts and graphs, but Power BI provides highly interactive and advanced dashboards.
- **Nature:** Excel is mostly used for manual data entry and calculation, while Power BI is focused on automated data flows and sharing.
- **Connection:** Excel has limited external data connections, but Power BI connects to hundreds of data sources.

Q2. Explain the concept of data modeling in Power BI.

Ans:

Data modeling is the process of connecting different tables using relationships so they can work together as a single source. It allows us to combine data from multiple tables to create meaningful reports. For example, connecting a 'Sales' table to a 'Products' table using a common Product ID.

Q3. What are the different types of connections available in Power BI?

Ans:

There are four main connection modes available for users:

1. **Import:** Data is uploaded directly into Power BI (best performance).
2. **DirectQuery:** Power BI queries the data directly from the source without importing it.
3. **Live Connection:** Used to connect directly to pre-defined Analysis Services models.
4. **Composite Model:** A mix of both Import and DirectQuery methods.

Q4. How do you handle data transformation in Power BI?

Ans:

Data transformation is done in the "Power Query Editor." It involves cleaning the data before using it. Common tasks include:

- Deleting empty or unnecessary rows and columns.
- Fixing data types (e.g., changing text to date).
- Replacing missing values.
- Merging or splitting columns for better organization.

Q5. What is DAX (Data Analysis Expressions) and why is it important in Power BI?

Ans:

DAX is a library of functions and operators used to create formulas in Power BI. It is important because it allows users to perform complex calculations, create custom measures, and build advanced business logic that is not possible with basic data.

Q6. Can you explain the difference between calculated columns and measures in Power BI?

Ans:

- **Calculated Column:** It adds a new column to your table and stores data in the model. It is calculated for every row immediately.
- **Measure:** It is a dynamic calculation that happens only when you use it in a visual. It does not take up extra space in the file.

Q7. How do you handle relationships between tables in Power BI?

Ans:

Relationships are managed in the "Model View" by dragging and dropping fields between tables. We use:

- **One-to-Many:** One record in a table matches many in another.
- **Cardinality:** Defining how tables are related.
- **Cross-filter direction:** Controlling how the data filtering flows between tables.

Q8. What is the purpose of a Power BI Gateway?

Ans:

A **Power BI Gateway** is a software application that acts as a **secure bridge** between the Power BI cloud service and your local data stored on a private network.

Its primary purpose is to allow the cloud to "reach back" into your office or home computer to **refresh data** and keep your reports up to date without compromising your network's security.

Q9. How can you schedule data refresh in Power BI Service?

Ans:

You can do this by going to the dataset settings in Power BI Service. Under the "Scheduled Refresh" section, you can set the specific time and frequency (daily or weekly) for the data to update automatically.

Q10. Explain the concept of row-level security (RLS) in Power BI.

Ans:

RLS is a security feature that limits what data a user can see based on their role. For example, a manager in New York can only see data related to the New York office, while others are restricted from seeing it.

Q11. What is the Power BI Desktop and how does it differ from Power BI Service?

Ans:

- **Power BI Desktop:** A Windows application used to create data models, transformations, and report designs locally.
- **Power BI Service:** An online cloud platform used to publish, share, and collaborate on those reports with others.

Q12. Explain the concept of Direct Query in Power BI.

Ans:

Direct Query means Power BI does not store the data internally. Instead, it retrieves data from the original source in real-time whenever a user interacts with the report. This is useful for very large databases.

Q13. What are Power BI templates and how are they useful?

Ans:

A template (.pbit) contains the report layout and data model structure but does not include the actual data. It is useful for sharing a report design that can be used with different datasets later.

Q14. How do you handle incremental data refresh in Power BI?

Ans:

Incremental refresh allows you to update only the newest data instead of reloading the entire dataset. This saves time and reduces the load on the system.

Q15. What is the role of Power Query in Power BI?

Ans:

Power Query is the engine used for the "Extract, Transform, and Load" (ETL) process. It helps in preparing and cleaning raw data before it is loaded into the Power BI model.

Q16. Difference between calculated columns and calculated tables.

Ans:

- **Calculated Column**: Adds a single new field to an existing table.
- **Calculated Table**: Creates a whole new table using a DAX formula (useful for creating date tables or summary tables).

Q17. How do you create custom visuals in Power BI?

Ans:

You can either download unique visuals from the Microsoft AppSource marketplace or create your own using the Power BI Visuals SDK and TypeScript.

Q18. Best practices for optimizing performance in Power BI.

Ans:

- Use the "Star Schema" for better table relationships.
- Remove any data or columns that are not needed for the report.
- Prefer using Measures over Calculated Columns whenever possible.
- Use simple visuals to keep the report loading fast.

Q19. Integration with other Microsoft products.

Ans:

Power BI integrates perfectly with Azure for data storage, Excel for deep analysis, and Microsoft Teams or SharePoint for easy sharing and collaboration.

Q20. Explain the concept of aggregations in Power BI.

Ans:

Aggregations are summarized versions of large tables. They improve performance because Power BI can show the summary (like total sum) instantly without scanning millions of individual rows.

Q21. How do you handle error handling and data quality?

Ans:

We use Power Query tools to identify errors in columns. Common steps include removing duplicates, fixing invalid data types, and using the "Replace Errors" feature to maintain data quality.

Q22. What is the purpose of Power BI Embedded?

Ans:

Power BI Embedded allows developers to put Power BI reports directly into their own custom websites or applications, so users can see the reports without logging into the Power BI website.